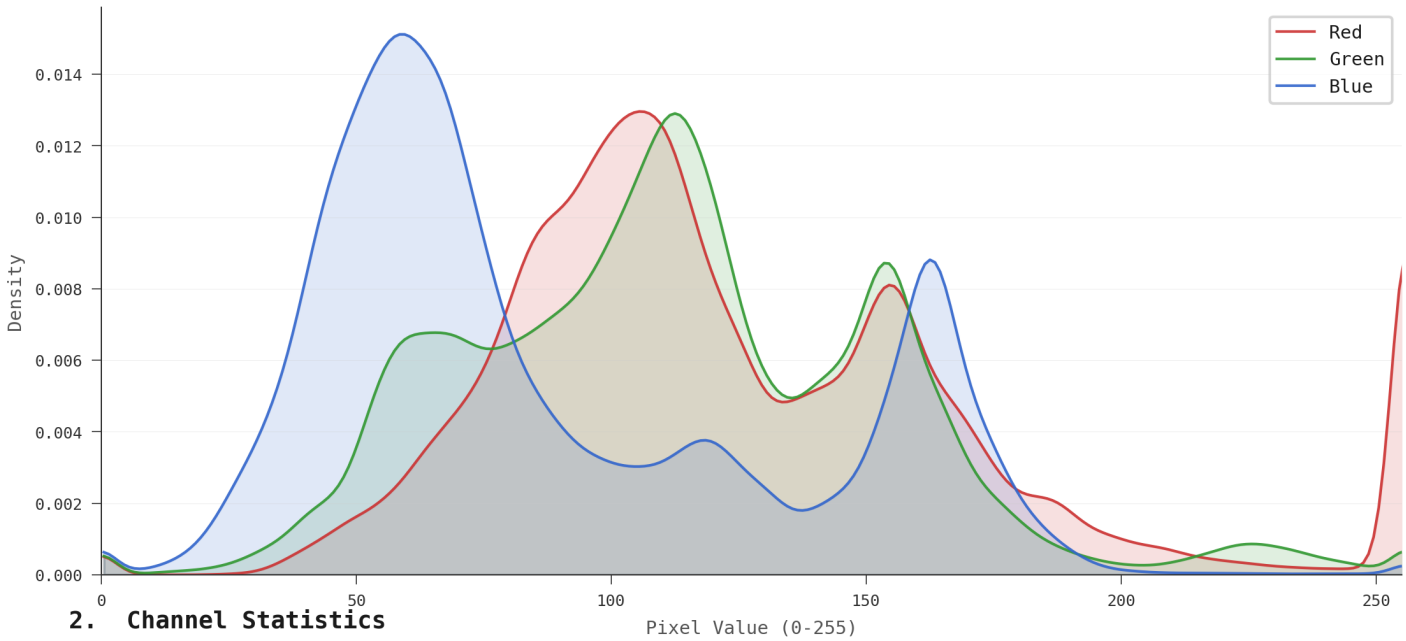
Reference Data Sheet: KODIM22

Subject: Red Barn and Fences

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	121.7830	44.5188	1981.9227	1.1508	0.9451	0	255
Green	112.4031	40.9399	1676.0753	0.4604	0.4691	0	255
Blue	88.6816	46.6028	2171.8180	-0.7035	0.7033	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.8714	0.8001
G	0.8714	1.0000	0.8752
B	0.8001	0.8752	1.0000

Pairwise Pearson Correlation Coefficients

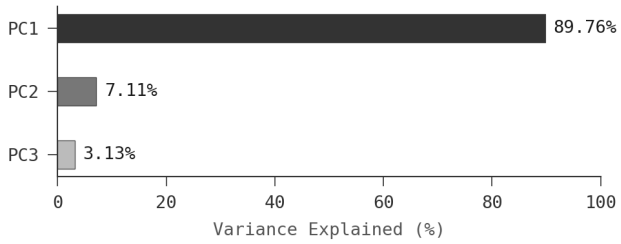
r(R-G)	=	**0.871394**
r(R-B)	=	**0.800057**
r(G-B)	=	**0.875171**
Avg |r|	=	**0.848874**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{1981.9277} & 1588.2027 & 1659.8820 \\
1588.2027 & \mathbf{1676.0796} & 1669.7540 \\
1659.8820 & 1669.7540 & \mathbf{2171.8235}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM22

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5769	-0.5438	**-0.6095**	5232.6152
PC2	-0.7151	-0.0243	0.6986	414.4790
PC3	0.3947	-0.8389	0.3748	182.7367

6. Derived Metrics

Condition Number

28.63

(l1/l3) PC1/PC2: 12.62 | PC2/PC3: 2.27

Blue Channel Independence

10.5%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Blue

Loading: -0.6095 | Blue dominant

Classification: Weakly One-Dimensional (PC1 = 89.76%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.955737	0.959898	0.957817
Green	0.943790	0.948469	0.946130
Blue	0.956775	0.963268	0.960022

8. Average Local Variance (3x3)

Channel	Value
Red	129.0350
Green	137.5900
Blue	130.4296

9. Redundancy Profile

Classification: Weakly One-Dimensional | Condition Number: 28.63 | PC1: 89.76%

Correlation: G–B pair leads at r = 0.875171 | R–G: 0.871394 | R–B: 0.800057

Blue Channel: 10.5% independent | Moderate decoupling from PC1

Spatial: Avg autocorrelation: 0.954656 | Avg local variance: 132.3515

Eigenvector: Blue dominant | B (-0.6095) leads | R (-0.5769), G (-0.5438)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."